Figure 7 Complete Energetic & Geometric Decomposition

Benchmarking Single-Site $+U_{\mathrm{Cr}}$ ($U_{\mathrm{Cr}} = 3.29$ eV) vs. Multi-Site $+U_{\mathrm{all}}$ ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV) & Pure PBE Baseline

Catalyst System	Site Motifs	$d_{\mathrm{TM-H}}$ (Å)		$E^{\mathrm{H}}_{\mathrm{int}}$ (eV)		E_{def} (eV)		ΔG_{H^*} (eV)			Sabatier Status
		$+U_{\mathrm{Cr}}$	$+U_{\mathrm{all}}$	$+U_{\mathrm{Cr}}$	$+U_{\mathrm{all}}$	$+U_{\mathrm{Cr}}$	$+U_{\mathrm{all}}$	PBE	$+U_{\mathrm{Cr}}$	$+U_{\mathrm{all}}$	
A. Surface-Adsorbed Coordination Family [Figure 7(a–c, g)]											
$CrCl_3$−Co (ads)	3-fold Cl (S_2)	1.55	1.55	−2.46	−2.40	0.18	0.23	+0.18	+0.22	+0.33	★ Optimal Sweet Spot
$CrCl_3$−Fe (ads)	3-fold Cl (S_2)	1.59	1.61	−2.35	−1.35	0.38	0.98	+0.32	+0.47	+2.07*	Active Window ($+U_{\mathrm{Cr}}$)
$CrCl_3$−Ni (ads)	3-fold Cl (S_2)	1.45	1.44	−1.49	−1.55	0.09	0.52	+0.81	+1.05	+1.42*	Moderate Overpotential
B. Interstitial Pore Embedded Coordination Family [Figure 7(d–f, h)]											
$CrCl_3$−Co (emb)	6-fold Cl (Pore)	1.47	2.38	−0.94	−0.92	0.21	0.20	+1.76	+1.77	+1.78	Strongly Endergonic
$CrCl_3$−Fe (emb)	6-fold Cl (Pore)	1.48	1.53	−1.67	−1.32	0.55	0.91	+1.35	+1.40	+2.11	Shielded Active Core
$CrCl_3$−Ni (emb)	6-fold Cl (Pore)	1.52	1.63	−0.82	−0.78	0.73	0.81	+1.85	+2.37	+2.49	Coordination Saturated

Physical Definitions & Notes:

• **Interaction vs. Deformation**: $\Delta G_{\mathrm{H}^*} = E^{\mathrm{H}}_{\mathrm{int}} + E_{\mathrm{def}} + \Delta E_{\mathrm{corr}} + \Delta E_{\mathrm{ref}}$, where $E^{\mathrm{H}}_{\mathrm{int}}$ is referenced to an isolated spin-polarized H atom ($\mathrm{H}^\bullet$, ISPIN = 2, −1.116 eV) (teal), E_{def} is the substrate distortion penalty (gold), $\Delta E_{\mathrm{corr}} = \Delta E_{\mathrm{ZPE}} - T\Delta S$ is the explicit vibrational correction (+0.18 to +0.26 eV), and $\Delta E_{\mathrm{ref}} = E(\mathrm{H}^\bullet) - \frac{1}{2}E(\mathrm{H_2}) = +2.265$ eV (≈ +2.26 eV) is the spin-polarized gas-phase dissociation reference connecting atomic H to molecular $\frac{1}{2}\mathrm{H_2(g)}$.

• **Sabatier Benchmark**: Co (ads) delivers the lowest overpotential of all single-atom configurations ($\Delta G_{\mathrm{H}^*} = +0.220$ eV under $+U_{\mathrm{Cr}}$, +0.330 eV under $+U_{\mathrm{all}}$), maintaining the most active HER site on $CrCl_3$.

• **Pore Relaxation Asterisk (*)**: Values for Fe and Ni under $+U_{\mathrm{all}}$ are referenced against their spontaneous deep-pore relaxed ground states ($\Delta z_{\mathrm{TM-Cr}} \approx 0.18$ Å). On the metastable surface hollow (S_2) site, ΔG_{H^*} remains at +0.47 eV (Fe) and +1.05 eV (Ni).